Roll No.: ....

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: B.Tech. *Semester: VIII*

Name of the Course: Satellite Communications *Course Code:* **TEC 851**

Time: 90 Minutes Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the su' questions***.
(ii) Each question carries 10 marks

Q1	(10 marks)	CO 1
(a)	Explain the concept of satellite communication. Elaborate on the various advantages of satellite communication.	
	OR	
(b)	A satellite is revolving in a circular orbit distance of 2620 km from the surface of the earth. The time period of the revolution of the satellite is (Radius of the earth =6380 km , mass of the earth $=6\times10^{24}$ kg, $G = 6.67\times10^{11} N-m^2/kg^2$	
Q2	(10 marks)	CO 1
(a)	Describe Kepler's laws of planetary motion and explain their significance in satellite communication and orbital mechanics.	
	OR	
(b)	The apogee and perigee of an elliptical satellite orbits are 3000 km and 200 km. Determine the eccentricity and semi-minor axis.	
Q3	(10 marks)	CO 1
(a)	Explain the various types of satellite orbits, including their advantages, disadvantages, and applications in communication systems.	
	OR	
(b)	The apogee and perigee of an elliptical satellite orbits are 4000 km and 300 km. Determine the semi-minor axis and semi-minor axis.	
Q4	(10 marks)	CO 2
(a)	Discuss the major subsystems of a satellite, focusing on their roles in ensuring proper operation and communication.	
	OR	
(b)	An artificial satellite revolves round the earth at a height of 1000km. The radius of the earth is 6.38×10^3km. Mass of the earth $=6\times10^{24}$kg; $G=6.67\times10^{-11}$ $Nm^2\ kg^{-2}$. Find the orbital speed of the satellite.	
Q5	(10 marks)	CO 2
(a)	What is Tracking, Telemetry, and Command (TTC and M), and how does it support satellite operations?	
	OR	
(b)	What is Spot Beam Technology, and how does it improve communication coverage and reliability?	